

0) Material Inventory and Review Scope

Sections received: Abstract, Introduction, Methods (2.1–2.5), Results (3.1–3.4), Discussion (4.0–4.2), Tables 1–2 (inline), Conflict of Interest, Author Contributions (incomplete), Funding, Acknowledgments, Data Availability Statement, References (1–13).

Sections/materials NOT received:

Supplementary materials (referenced at end of manuscript: "Full prompt templates for all experimental conditions and detailed criterion-level statistical tables are available as Supplementary Material")

Figures (none appear in this manuscript version)

Author Contributions section is incomplete ("Need to insert")

Review confidence: Limited — Supplementary materials containing prompt templates and criterion-level statistical tables are not available for verification. Claims about criterion-level analyses (Section 3.2) cannot be fully audited without the supplementary tables.

1. Overall Recommendation

Major Revision

The manuscript addresses a timely question and has a logical experimental framework, but severe pseudoreplication inflating effective sample sizes, inappropriate use of independent-samples t-tests on non-independent data, and overclaiming from a 3-application study constitute major threats to the validity and interpretability of the reported findings.

2. Summary Assessment

This study compares three prompt engineering strategies (zero-shot, few-shot, few-shot + strict) across three commercial LLMs reviewing three de-identified seed grant applications, benchmarked against 12 human expert reviews. The experimental framework is logical and the data contamination control (excluding the training application from Experiments 2–3) is appropriate. However, the statistical analysis treats pseudoreplicates (3 iterations from the same LLM on the same application) as independent observations, substantially inflating apparent sample sizes and invalidating the reported inferential statistics. The effective sample is 3 applications — insufficient to support the generalizability claims made. The manuscript makes several interpretive claims (e.g., "literal compliance bias," prioritizing prompt optimization over model selection) that exceed what 3 applications and 63 LLM reviews with clustered structure can support.

3. Fatal Flaws

No single fatal flaw warranting outright rejection was identified, but the pseudoreplication and statistical independence violations described under Major Concerns approach fatal severity if not addressed, as they undermine all inferential results.

4. Major Concerns

4.1 Pseudoreplication and inflated sample sizes

Issue: Three iterations per vendor-application-experiment combination are treated as independent observations (e.g.,  $n = 27$  LLM reviews in Experiment 1). These iterations are pseudoreplicates — three outputs from the same model on the same input. They are not independent in the same sense as

distinct human reviewers evaluating the same application.

Where it appears: Methods 2.4, all Results sections, Tables 1–2.

Why it matters: Treating pseudoreplicates as independent observations artificially inflates degrees of freedom, narrows standard errors, and produces unreliable p-values and effect sizes. The effective LLM sample size per experiment is closer to 9 (3 vendors × 3 applications) or even 3 (applications) depending on the analysis level, not 18–27.

What would strengthen it: Use mixed-effects models or hierarchical analyses that account for nesting (iterations within vendor-application combinations, scores within applications). Alternatively, average iterations within each vendor-application cell before analysis. Report both the inflated and corrected analyses.

#### 4.2 Violation of independence assumption in t-tests

Issue: Independent-samples t-tests are used to compare LLM vs. human scores, but scores are clustered within applications. Multiple LLM and human scores evaluate the same 2–3 applications, creating within-application correlation that t-tests cannot account for.

Where it appears: Methods 2.5, Results 3.1.

Why it matters: Ignoring clustering within applications violates the independence assumption, inflating Type I error. The ANOVA ( $F(2,60) = 27.45$ ) is similarly affected.

What would strengthen it: Use mixed-effects models with application as a random effect (or at minimum, a fixed effect). Report intraclass correlation coefficients to characterize clustering.

#### 4.3 Extremely small effective sample size ( $n = 3$ applications)

Issue: All findings are based on reviews of 3 grant applications (reduced to 2 in Experiments 2–3). This is acknowledged in the limitations but insufficiently reflected in the strength of the claims.

Where it appears: Throughout, but particularly Abstract, Discussion 4.0.

Why it matters: With 2–3 applications, the study cannot distinguish prompt engineering effects from application-specific effects. An LLM might score better on one application than another for idiosyncratic reasons unrelated to prompt strategy. The apparent "15.2-point swing" attributed to prompt engineering could be confounded with application characteristics.

What would strengthen it: Frame all findings explicitly as case-study-level evidence. Remove or heavily qualify claims about what "should be prioritized" or what practitioners "should" do. The abstract conclusion ("LLMs can approximate human-level scoring when calibrated with diverse examples") requires substantial qualification.

#### 4.4 Training data limited to a single application

Issue: Few-shot training examples in Experiments 2–3 consist of four human reviews of a single application (FC3). The LLM learns scoring patterns from one application's reviews, which may not generalize.

Where it appears: Methods 2.3 (Experiments 2–3).

Why it matters: The improved alignment in Experiment 2 could reflect the LLM calibrating to FC3's specific score range (68–92) rather than learning generalizable scoring behavior. With reviews from only one training application, the few-shot approach is confounded with application-specific calibration.

What would strengthen it: Acknowledge this as a design limitation explicitly. Discuss whether the score range of 68–92 in training examples may have anchored LLM outputs regardless of test application quality.

#### 4.5 Multiple comparisons without correction

Issue: The study conducts numerous hypothesis tests: 3 t-tests (one per experiment), 6+ criterion-level comparisons per experiment, vendor-level comparisons, and one ANOVA, all at  $\alpha = 0.05$  without correction.

Where it appears: Methods 2.5, Results 3.1–3.3.

Why it matters: With at least 20+ statistical tests, the family-wise error rate substantially exceeds 0.05. The criterion-level findings (e.g., External Funding Potential  $p = 0.004$ , Innovation and Impact  $p = 0.003$ ) may not survive correction.

What would strengthen it: Apply Bonferroni, Holm, or FDR correction, or explicitly state that criterion-level analyses are exploratory with no multiplicity adjustment. Indicate which findings survive correction.

#### 5) Moderate Concerns

##### 5.1 Cohen's d formula unspecified with unequal variances

Issue: LLM standard deviations (~5.3–5.8) are roughly 2.5× smaller than human standard deviations (~13.3–13.7). The pooled-SD formulation of Cohen's d is sensitive to this variance heterogeneity.

Where it appears: Methods 2.5, Table 1.

Why it matters: The choice between pooled-SD, control-group SD, or Glass's delta substantially affects effect size estimates when variances differ this markedly.

What would strengthen it: Specify which d formula was used. Consider reporting Glass's delta using the human (control) group SD as the reference, and report confidence intervals for effect sizes.

##### 5.2 "Broad agreement" metric is uninformative

Issue: "Broad agreement" (positive vs. negative funding direction) is reported at 33.3–50.0%, but given that both LLMs and humans overwhelmingly provide positive recommendations, this metric has a very high expected agreement by chance alone.

Where it appears: Results 3.4, Table 1.

Why it matters: Without chance-corrected agreement (e.g., Cohen's kappa), the 33.3–50.0% broad agreement rates are uninterpretable. The metric does not distinguish systematic agreement from base-rate agreement.

What would strengthen it: Report Cohen's kappa or similar chance-corrected agreement statistic for both broad and exact categories.

##### 5.3 Vendor-level comparisons not inferentially tested

Issue: Table 2 presents vendor mean scores and deviations, but no inferential statistics are provided for vendor comparisons. Claims about xAI Grok achieving "near-perfect alignment" or Google Gemini demonstrating "effective recalibration" are based solely on descriptive values.

Where it appears: Results 3.3, Table 2.

Why it matters: Without tests or confidence intervals, vendor-level claims are anecdotal. The differences between vendors could be within sampling variability.

What would strengthen it: Either provide inferential statistics for vendor comparisons or explicitly label all vendor-level observations as descriptive.

##### 5.4 LLM parameter details omitted

Issue: The manuscript states "all used standardized parameters" without specifying temperature, top-p, max tokens, system prompts, or other API settings.

Where it appears: Methods 2.4.

Why it matters: Temperature and sampling parameters directly affect output variability and are critical for reproducibility. The observed "lower variability" of LLMs could partially reflect low-temperature

settings rather than inherent consistency.

What would strengthen it: Report all API parameters. If temperature was set to 0 or near 0, discuss this as a contributor to the observed lower variability.

#### 5.5 Lower LLM variability may not be advantageous

Issue: The 49–59% reduction in score variability is presented as a strength ("higher inter-rater consistency"). However, lower variability could indicate range restriction — LLMs failing to differentiate between applications of genuinely different quality.

Where it appears: Abstract, Results 3.4, Discussion.

Why it matters: If LLMs compress scores toward a central tendency (as suggested by the "Fund with Revisions" preference), reduced variability may reflect insensitivity to application quality rather than superior consistency.

What would strengthen it: Discuss range restriction as an alternative interpretation. Report whether LLMs differentiated between the 3 applications in the same rank order as humans.

#### 5.6 Model version reporting insufficient

Issue: Models are described as "September 2025 versions" of GPT-4o, Gemini 1.5 Flash, and Grok-beta. Exact version strings or API snapshot identifiers are not provided.

Where it appears: Methods 2.4.

Why it matters: LLM behavior can change with minor version updates. Without precise version identifiers, exact replication is not possible.

What would strengthen it: Report the exact model identifiers used at the API level (e.g., gpt-4o-2024-08-06).

### 6) Minor Concerns

#### 6.1 Author Contributions section incomplete

Issue: Section reads "Need to insert."

Where it appears: Author Contributions.

Why it matters: Incomplete for submission. Minor but indicates the manuscript is not submission-ready.

#### 6.2 Inconsistent terminology: "repeated-measures comparative evaluation"

Issue: The study is described as "repeated-measures" (Methods 2.1), but this is not a repeated-measures design in the standard statistical sense. There is no within-subject factor where the same units are measured under multiple conditions.

Where it appears: Methods 2.1.

Why it matters: May mislead readers about the study design and the appropriateness of the statistical methods.

What would strengthen it: Use a more precise design descriptor (e.g., "multi-condition, multi-vendor comparative evaluation").

#### 6.3 Reference 9 may be misattributed

Issue: Reference 9 is attributed to "Bai Y, Kadavath S, Kundu S, Askell A, Kernion J, Jones A, et al." with the title "Large Language Models for Scientific Peer Review." The Bai et al. author list closely matches publications on Constitutional AI / RLHF from Anthropic, not peer review. This reference should be verified.

Where it appears: References.

Why it matters: If the citation is incorrect, the claim it supports (importance of prompt design) may be

unbacked.

#### 6.4 "Brief report" framing

Issue: The manuscript is formatted as a brief report but contains substantial methodological detail and multiple experiments. The framing may constrain appropriate space for addressing the statistical and interpretive issues raised.

Where it appears: Overall structure.

Why it matters: Some concerns (e.g., fuller reporting of model parameters, mixed-effects models, corrected agreement statistics) may be difficult to address within brief-report word limits.

#### 7) Internal Consistency Audit

The following numerical checks were performed:

Total LLM reviews:  $27 + 18 + 18 = 63$ . ✓ Matches abstract.

Human reviews per experiment: 12 (Exp 1), 8 (Exp 2), 8 (Exp 3). ✓ Consistent with design.

Mean differences from Table 1:

Exp 1:  $83.63 - 79.08 = +4.55$  ✓

Exp 2:  $80.56 - 83.13 = -2.57$  ✓

Exp 3:  $72.44 - 83.13 = -10.69$  (Table reports  $-10.68$ ; rounding within  $\pm 0.01$ )

Table 2 vendor means averaging to Table 1 LLM means:

Exp 1:  $(88.44 + 77.33 + 85.11) / 3 = 83.63$  ✓

Exp 2:  $(82.33 + 76.00 + 83.33) / 3 = 80.55$  (Table 1 reports 80.56; rounding within  $\pm 0.01$ )

Exp 3:  $(78.00 + 69.83 + 69.50) / 3 = 72.44$  ✓

Table 2 deviations from human mean: All check within  $\pm 0.01$  rounding tolerance.

Discussion "15.2-point swing":  $+4.55$  to  $-10.68 = 15.23 \approx 15.2$  ✓

Discussion "7.3-point spread": Matches Exp 2 vendor range ( $83.33 - 76.00 = 7.33$ ). Note: this cherry-picks Exp 2; Exp 1 vendor spread is 11.11 and Exp 3 is 8.50. The claim that "prompt engineering effects exceed vendor differences" holds for Exp 2 but is not consistently true across experiments.

ANOVA df:  $F(2,60)$  implies  $N = 63$ , treating all LLM reviews as independent. ✓ Consistent with reported  $n$  but reflects the pseudoreplication issue.

One inconsistency of note: The Discussion states "prompt engineering effects exceed vendor differences (7.3-point spread)" as a general claim, but this is true only for Experiment 2. In Experiment 1, the vendor spread (11.11 points) exceeds the prompt effect for that condition ( $+4.55$ ). This selective comparison overstates the generality of the finding.

#### 8. Statistical Review (Simple-First)

Denominators: Defined (number of reviews per group). However, the unit of analysis is problematic — reviews are treated as independent when they are nested within applications and vendors.

Missing data: Not reported. It is unclear whether any LLM calls failed or produced unparseable output. No mention of exclusions from the 63 planned reviews.

Effect sizes: Cohen's  $d$  is reported but without confidence intervals and without specifying the formula used (critical given  $\sim 2.5\times$  variance ratio between groups).

Confidence intervals: Not reported for any comparison.

Multiplicity: At least 20+ tests are conducted without correction.

Clustering/non-independence: Not addressed. This is the most consequential statistical concern.

Scores are nested within applications (3 apps), within vendors (3 LLMs), and within iterations (3 repeats). Independent-samples  $t$ -tests and one-way ANOVA ignore all three clustering levels.

Would a simpler analysis be more appropriate? Yes. Given only 3 applications, a descriptive analysis with appropriate uncertainty characterization (e.g., bootstrap confidence intervals accounting for clustering, or application-level summary statistics) would be more honest than inferential statistics that depend on independence assumptions.

Primary endpoint clarity: The primary endpoint (total score out of 100) is clearly defined.

#### 9. Supplementary Materials Assessment

Supplementary materials were not provided for review. The manuscript references them at the end: "Full prompt templates for all experimental conditions and detailed criterion-level statistical tables are available as Supplementary Material."

Impact of missing supplements:

Prompt templates are essential for evaluating whether Experiment 3's "strict instructions" were reasonable or unusually strong — the brief quotes in Methods 2.3 ("Apply high standards," "Reserve high scores for truly exceptional proposals," "Be critical in your assessment") provide only a partial view. Criterion-level statistical tables underpin the findings reported in Results 3.2. Without these tables, the specific p-values and effect magnitudes cited for criterion-level analyses cannot be verified. This is classified as a Major Concern — key claims depend on supplementary data that cannot be audited.

#### 10) Claims That Overreach the Evidence

Abstract: "LLMs can approximate human-level scoring when calibrated with diverse examples" — Based on 2 test applications in one experimental condition at one institution reviewing \$5,000 seed grants.

"Can approximate" implies a general capability conclusion that the evidence does not support.

Suggested direction: "In this pilot, few-shot prompting produced LLM scores not significantly different from human scores for two seed grant applications."

Abstract: "49–59% lower score variability" presented as a finding of LLM consistency — Could equally indicate range restriction or temperature-dependent output compression. Suggested direction:

Acknowledge both interpretations.

Discussion: "prompt optimization should be prioritized over model selection" — Based on comparing 3 prompt conditions with 3 vendors across 3 applications. The vendor spread exceeded the prompt effect in Experiment 1 (11.1 vs. 4.6 points). Suggested direction: "In this sample, prompt condition explained more score variance than vendor selection in the best-aligned experiment, though this pattern was not consistent."

Discussion: "likely reflecting a literal compliance bias where LLMs prioritize explicit instructions over the calibrating influence of training examples" — This is a mechanistic attribution based on behavioral observation of 18 reviews. Suggested direction: Frame as speculation or hypothesis, not as a likely explanation.

Discussion 4.2: "Institutions should conduct context-specific validation before deployment, use multiple diverse training examples for calibration without directive language" — Advisory language ("should") derived from a 3-application pilot is premature. Suggested direction: Frame as hypotheses for future testing rather than recommendations.

#### 11. Priority Revisions Before Reconsideration

Address pseudoreplication and non-independence. Either re-analyze with mixed-effects models

(application and vendor as random/fixed effects) or re-frame all analyses as descriptive with appropriate uncertainty characterization. This is the most critical revision — all inferential results depend on it.

Substantially temper claims to match evidence. Rewrite abstract conclusions, discussion implications, and practitioner recommendations to reflect case-study-level evidence from 3 applications at a single institution. Remove "should" language for institutional practice.

Provide full LLM parameters. Report temperature, top-p, max tokens, and exact model version identifiers. Discuss whether low-temperature settings contribute to the observed lower variability.

Address multiplicity. Apply correction for multiple comparisons or explicitly designate all criterion- and vendor-level analyses as exploratory.

Report chance-corrected agreement. Replace or supplement "broad agreement" with Cohen's kappa or equivalent.

Provide supplementary materials for review. Prompt templates and criterion-level tables are necessary for full evaluation.

Acknowledge single-application training limitation. Discuss the confound between general scoring calibration and application-specific anchoring in Experiments 2–3.